

Interação Pessoa-Máquina

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Sous-Chef Ratchet: Kitchen Assistant

Stage 1: Project Proposal



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Problem Description

People who cook at home often have difficulty keeping track of the ingredients available in their kitchen, their quantities and expiration dates. This can make deciding what to cook somewhat difficult, lead to buying products that are already available at home, and contribute to food waste when ingredients are unused or even forgotten before they expire.

The cooking process itself also presents usability challenges. Users frequently need to consult recipes while their hands are occupied, wet or dirty, forcing them to interrupt the activity to interact with a mobile device. Furthermore, information about available ingredients, recipes and shopping needs is often managed separately, requiring users to switch between different tools or rely on memory.

Target Users

The primary target users are university students and young adults who live independently or in shared households and cook at home regularly or occasionally. In particular, the project targets non-expert cooks who use mobile devices in their daily lives and would benefit from assistance in managing ingredients, choosing meals and following recipes.

Project Goal

The goal of the project is to design and evaluate a mobile kitchen assistant that supports users throughout the process from managing their available food to preparing a meal.

A possible solution would allow users to maintain a kitchen inventory, discover recipes according to the ingredients they already have, identify missing or products about to expire, and transfer missing ingredients directly to a shopping list.

During meal preparation, the application could provide a dedicated cooking mode with simplified step by step instructions, adaptable controls, timers and optional voice interaction, reducing the need to touch the device while cooking.

The specific functionality and interaction mechanisms may be modified throughout the project according to the results of user analysis and usability evaluation.

Competing Application —

KitchenPal is an existing kitchen-management application with functionality related to the proposed project, including pantry management, expiration tracking, recipe recommendations and shopping-list support.

A positive aspect of its design is the integration of several activities related to food management into a single application, allowing information about stored ingredients to be reused in other tasks such as recipe discovery and shopping.

However, its broad range of features may result in a management-oriented experience with considerable amounts of information and functionality. Following recipes may also be unnecessarily hard, since they lead to 3rd party sites. Our project intends to explore a more task and context oriented interaction, particularly the transition from deciding what to cook to actually preparing the meal. Special attention will be given to designing a simplified cooking interface suitable for situations in which the user's attention and hands are occupied.

(Supercook, Paprika)